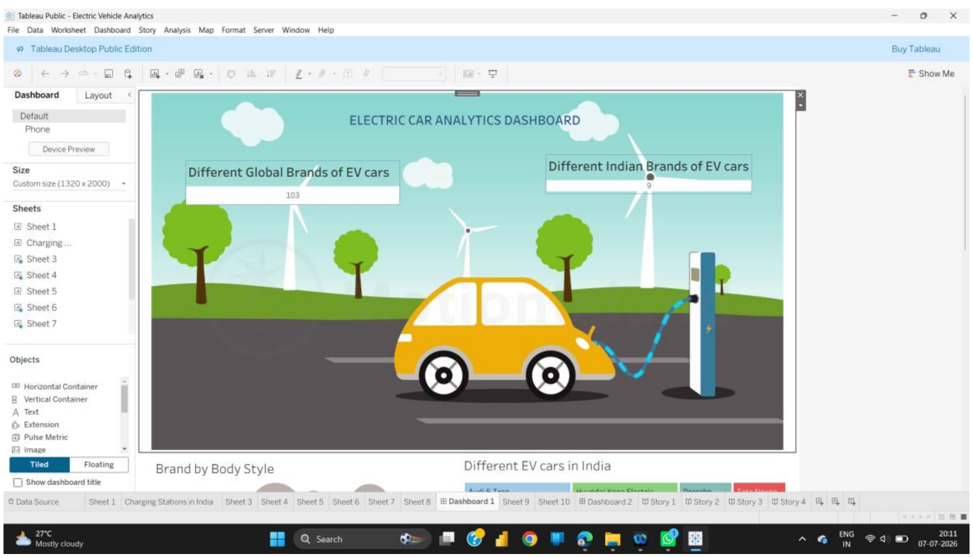
PROJECT REPORT

Electric Vehicle Charge and Range Analysis

A Data Analytics & Visualization Project using Tableau



Submitted in partial fulfilment of the requirements of the Internship Programme

Submitted by:

Poojitha Attili - attilipooja@gmail.com
Sravanthi Kanisetti - kanisettisravanthi@gmail.com
Naga Venkata Sasi Kumar Avala - anvsasikumar19@gmail.com
Mohan Chukka - chnohan022@gmail.com
Sruthi Sahithi Bandaru - sruthisahithi118@gmail.com

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Project Overview

Electric Vehicles (EVs) are rapidly emerging as the future of personal and public transportation, driven by the urgent need to reduce carbon emissions, dependency on fossil fuels, and rising fuel costs. As the EV market expands globally and in India, consumers and stakeholders are faced with an increasing number of vehicle choices, each differing in range, charging time, efficiency, price, and performance. At the same time, the success of EV adoption depends heavily on the availability of a reliable charging infrastructure.

The project titled "Electric Vehicle Charge and Range Analysis" is a data analytics and visualization solution built using Tableau. It analyzes and visualizes electric vehicle specifications — such as range, top speed, acceleration, efficiency, battery capacity, and price — across global and Indian EV markets, and maps the distribution of EV charging stations across India. The project integrates four real-world datasets and presents the insights through a set of interactive dashboards and a guided story, enabling users to compare vehicles and understand charging accessibility at a glance.

This project was developed as part of an internship program, with the objective of applying data analytics, data visualization, and business intelligence skills on a real-world, sustainability-focused domain.

1.2 Purpose

The primary purpose of this project is to provide a consolidated analytical view of the electric vehicle ecosystem so that prospective buyers, researchers, and policymakers can make informed decisions. Specifically, the project aims to:

- Compare electric vehicles across brands based on range, top speed, acceleration, efficiency, and price.
- Identify the most efficient and best-performing EV brands and models.
- Visualize the geographic distribution of EV charging stations across India by region, operator, and charger type.
- Highlight the diversity of EV body styles and models available in the Indian market.
- Support data-driven decision-making for EV purchase and charging-infrastructure planning.

By combining specification data with charging-infrastructure data in a single set of dashboards, the project bridges the gap between 'which EV to buy' and 'where can I charge it' — two of the most common concerns among prospective EV owners.

2. IDEATION PHASE

2.1 Problem Statement

Prospective EV buyers currently face two major challenges. First, EV specification data (range, efficiency, charging speed, price, top speed) is scattered across manufacturer websites and third-party portals, making side-by-side comparison difficult and time-consuming. Second, even after choosing a vehicle, buyers and existing EV owners often have limited visibility into the availability, location, and type of charging stations near them, which creates 'range anxiety' and slows EV adoption.

Problem Statement: "How might we help electric vehicle buyers and owners easily compare EV models on key performance parameters and understand the availability of charging infrastructure, so that they can make confident, informed decisions about purchasing and using an EV?"

2.2 Empathy Map Canvas

An empathy map was created to understand the perspective of a typical EV buyer/owner persona:

Category	Insights
Says	"I don't know which EV gives the best range for the price." / "Will I find a charging station on my route?"
Thinks	Worried about running out of charge (range anxiety); confused by too many brands and specifications; concerned about resale and running cost.
Does	Searches multiple websites and forums to compare EVs; visits showrooms; asks other EV owners; checks maps for charging stations near home/office.
Feels	Overwhelmed by scattered information; anxious about charging availability; excited about switching to sustainable transport but hesitant due to lack of clarity.

2.3 Brainstorming

The following ideas were brainstormed and evaluated to arrive at the final solution:

- An interactive dashboard comparing EV brands on range, efficiency, top speed, and acceleration.
- A geographic map showing charging station locations across India, colour-coded by operating region.
- A breakdown of charging connector/type distribution (AC, DC, CCS/CHAdeMO) by region.
- A price comparison chart for EVs available in the Indian market.
- A brand-wise and body-style-wise distribution view (SUV, Sedan, Hatchback, etc.) to show market variety.

- A guided 'Story' in Tableau that walks a first-time user through charging infrastructure, EV models, pricing, and brand distribution step-by-step.

After evaluating feasibility, data availability, and value to the end user, these ideas were consolidated into a single Tableau workbook containing multiple worksheets, two dashboards, and a four-part story.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	Touchpoint / Need
Awareness	Learns about EVs as an alternative to fuel vehicles	News, ads, word of mouth
Research	Searches for EV models, specifications and prices	EV specification dashboard
Comparison	Compares range, efficiency, top speed and price across brands	Brand comparison charts, price chart
Decision	Shortlists 2–3 models that fit budget and range needs	Story point walkthroughs
Charging Planning	Checks charging station availability in their region	Charging station map dashboard
Purchase & Use	Buys the EV and uses the charging network	Ongoing reference to dashboard

3.2 Solution Requirement

Functional Requirements:

- Import and connect multiple CSV datasets (global EV specs, Indian EV specs, charging station list) into Tableau.
- Provide visual comparison of EV brands on efficiency (Wh/km), top speed (km/h), and price.
- Display distinct counts of EV models available per brand and body style.
- Plot charging stations on a map of India using latitude/longitude, coloured by operating region.
- Show charger-type distribution (AC-001, DC-001, CCS/CHAdeMO) grouped by region.
- Combine individual worksheets into interactive dashboards and a navigable story.

Non-Functional Requirements:

- Usability — dashboards should be intuitive for non-technical users.
- Performance — visuals should render quickly even with 100+ records per sheet.
- Scalability — the workbook should support addition of new EV models or charging stations without redesign.
- Portability — published via Tableau Public so it can be accessed through a web browser.

3.3 Data Flow Diagram

The data flow for this project follows a straightforward analytics pipeline: raw CSV data sources are cleaned and loaded into Tableau, relationships/joins are established between tables where required, calculated fields are derived, and the resulting data model feeds the worksheets, which are then combined into dashboards and a story.

Stage	Description
1. Data Sources	ElectricCarData_Clean.csv, EVIndia.csv, Cheapestelectriccars-EVDatabase.csv, electric_vehicle_charging_station_list.csv
2. Data Preparation	Cleaning column headers, handling nulls, standardizing units (km/h, Wh/km, Lakhs/Euro)
3. Data Modelling	Loading tables into Tableau's data pane (EVIndia+ data source); establishing fields such as Brand, Car, region, PriceRange, Efficiency
4. Analysis / Worksheets	Sheet 1–Sheet 10: charging station counts, brand distribution, price chart, top-speed chart, efficiency chart, EV catalogue
5. Dashboards	Dashboard 1 (Electric Car Analytics Dashboard) and Dashboard 2 combine multiple sheets with filters and actions
6. Story	Story 1–Story 4 present a guided narrative: Charging Stations in India → Charging Station by Region and Type → Price of EVs → Brands and Models
7. End User	Views published dashboard/story on Tableau Public via web browser to compare EVs and locate charging stations

3.4 Technology Stack

Layer	Technology Used
Data Source Format	CSV (Comma Separated Values)
Data Visualization / BI Tool	Tableau Desktop (Public Edition)
Mapping Engine	Mapbox / OpenStreetMap (built into Tableau)
Publishing Platform	Tableau Public
Datasets	Global EV Specification Data, Indian EV Data, Cheapest EV Database, EV Charging Station List (India)
Supporting Tools	Microsoft Excel (data cleaning), Web Browser (for published dashboard access)

4. PROJECT DESIGN

4.1 Problem – Solution Fit

Problem	Solution Offered
Scattered EV specification data across brands	Consolidated Brand vs Efficiency, Top Speed, and Price charts in a single dashboard
Difficulty identifying the most efficient / fastest EVs	'Top 10 Most Efficient Brands' and 'Top Speed for Different Brands' visualizations
Lack of visibility into Indian EV market variety	Treemap of brand distribution and catalogue view of all Indian EV models by body style
Uncertainty about charging station availability	Interactive India map plotting charging stations coloured by operating region (NDMC, CMRL, Maha Metro, etc.)
No single source combining vehicle + infrastructure data	Unified Tableau workbook with cross-linked dashboards and a guided story

4.2 Proposed Solution

The proposed solution is an interactive Tableau analytics workbook, "Electric Vehicle Analytics", consisting of ten worksheets, two dashboards, and a four-part story. It draws data from four datasets covering 103 global EV models (33 brands), 12 Indian EV models, 180 entries from a cheapest-EV database, and 202 EV charging station records spanning 8 operating regions in India (NDMC, CMRL, Maha Metro, Noida Authority, SDMC, NKDA, NRANVP, and ANERT).

Key features of the proposed solution include:

- Distribution of Brands – a treemap showing the number of models offered by each global brand (Tata, Audi, BMW, BYD, Hyundai, Jaguar, Mercedes-Benz, Porsche, MG).
- Top 10 Most Efficient Brands – a bar chart ranking brands by maximum efficiency (Wh/km), led by Mercedes, Audi, and Tesla.
- Top Speed for Different Brands – a bar chart comparing maximum top speed (km/h) across 33+ brands, with Tesla topping at over 400 km/h.
- Price of Electric Cars – a horizontal bar chart of Indian EV price ranges (in Lakhs), from the Tata Tigor EV (~₹13.6L) to the Audi E-Tron GT (~₹180L).
- Brand by Body Style – a packed-bubble chart mapping brands against body styles (SUV, Sedan, Hatchback, MPV, etc.).
- Charging Stations in India – a map plotting the exact location of 202 charging stations, coloured by the managing authority/region.

- Charging Station by Region and Type – a stacked bar chart showing AC-001, DC-001, and CCS/CHAdeMO charger counts by region.
- Different EV Cars in India – a full catalogue view of all 12 Indian EV models available.

4.3 Solution Architecture

The solution follows a layered analytics architecture: a Data Layer (four CSV sources), a Data Modelling Layer (Tableau data source 'EVIndia+' with defined fields and relationships), a Visualization Layer (individual worksheets, Sheet 1 through Sheet 10), a Presentation Layer (Dashboard 1 – Electric Car Analytics Dashboard and Dashboard 2), and a Narrative Layer (Story 1 to Story 4, guiding the user from charging infrastructure to vehicle comparison). The published output is hosted on Tableau Public, making it accessible through any web browser without requiring the end user to install Tableau.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed over a series of sprints during the internship, following an iterative approach — starting from problem identification and data collection, moving through data preparation and dashboard development, and finishing with testing, story-building, and publishing.

Sprint	Activities Performed	Duration
Sprint 1	Problem statement finalization, empathy mapping, brainstorming, dataset identification and collection	Week 1
Sprint 2	Data cleaning and preparation (EV specs, charging stations, Indian EV data); requirement analysis and technology stack finalization	Week 2
Sprint 3	Building core worksheets – brand distribution, efficiency ranking, top speed comparison, price comparison, body style analysis	Week 3
Sprint 4	Building the charging station map and region-wise charger type analysis; assembling Dashboard 1 and Dashboard 2	Week 4
Sprint 5	Building the guided Story (Story 1–Story 4); functional and performance testing; bug fixing	Week 5
Sprint 6	Publishing to Tableau Public, final review, documentation and report preparation	Week 6

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing was carried out to ensure that all worksheets, dashboards, and the story rendered correctly, that filters and actions worked as expected, and that the workbook performed well given the dataset sizes involved (up to 202 rows per sheet).

Test Case	Description	Expected Result	Status
Data Connection	Verify all four CSV sources load without errors into the EVIndia+ data source	All fields available in the Data pane	Pass
Chart Rendering	Verify bar charts, treemaps, bubble charts and maps render with correct data	Charts display accurate values matching source data	Pass
Map Accuracy	Verify charging station coordinates plot at correct locations in India	202 stations plotted; 1 null location flagged	Pass
Filter Interaction	Apply region and brand filters on dashboards	Visuals update dynamically based on filter selection	Pass
Dashboard Load Time	Measure time to render Dashboard 1 with 4 combined sheets	Loads within acceptable time (< 3 seconds) on Tableau Public	Pass
Story Navigation	Navigate through Story 1 to Story 4	Each story point transitions smoothly and shows correct captions	Pass
Cross-Device View	Preview dashboard using Device Preview (Default / Phone)	Layout adapts appropriately for phone view	Pass

7. RESULTS

7.1 Output Screenshots

The following screenshots, captured from the published Tableau workbook (Tableau Public – Electric Vehicle Analytics), showcase the dashboards, worksheets, and story points developed as part of this project.

Electric Car Analytics Dashboard – Cover

The landing view of Dashboard 1, showing the overall theme and quick counts of global (103) and Indian (9) EV brands.

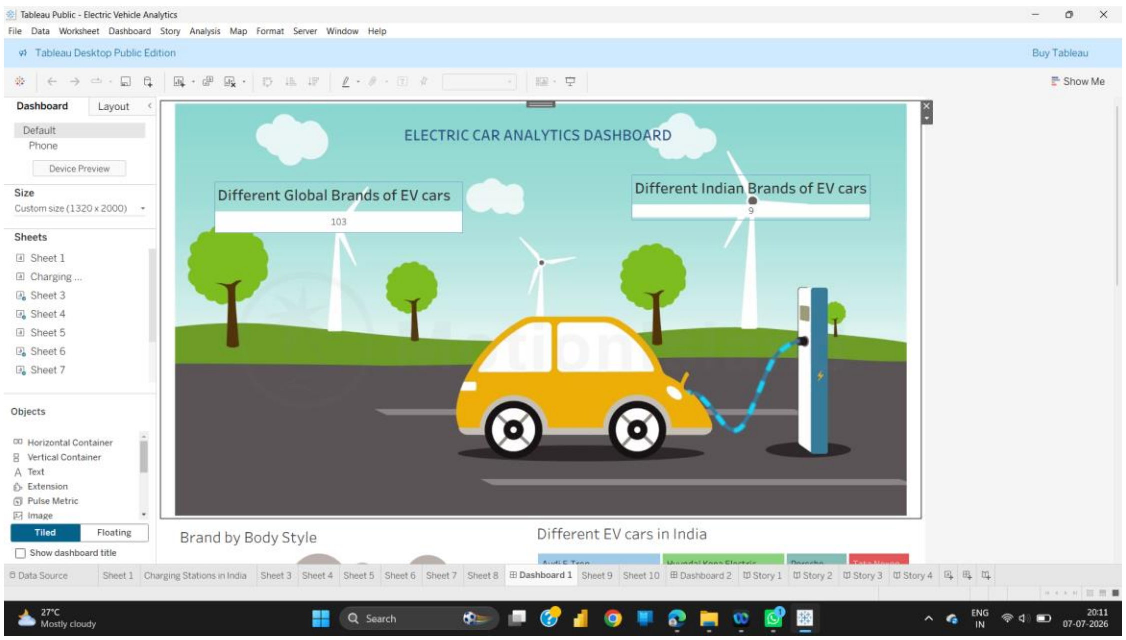


Figure: Electric Car Analytics Dashboard – Cover

Dashboard 1 – Brand by Body Style & EV Catalogue

Combined view showing the packed-bubble Brand-by-Body-Style chart alongside the catalogue of Indian EV models and brand distribution.

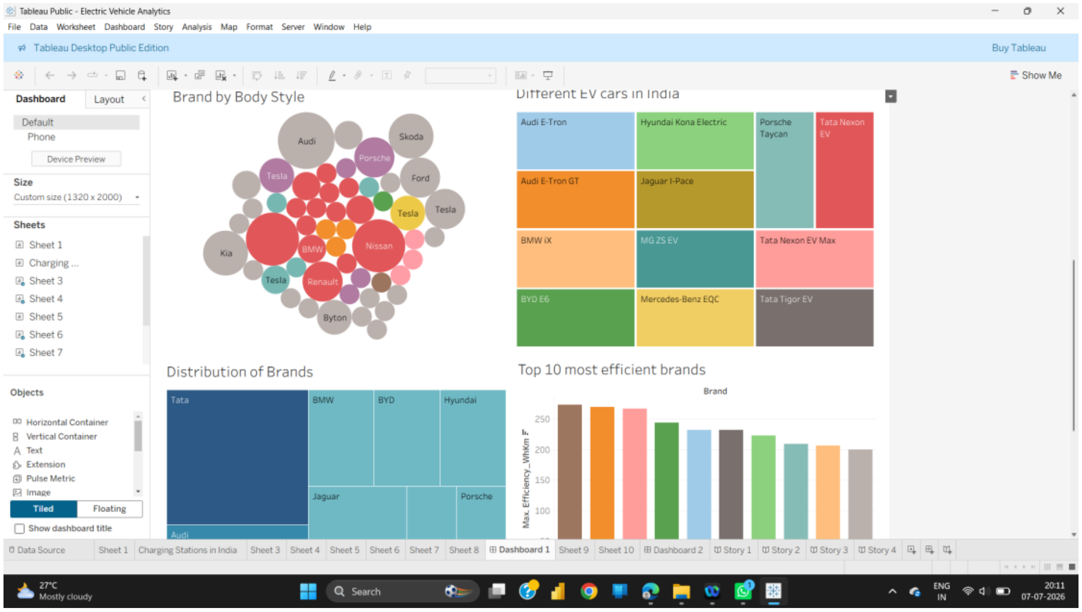


Figure: Dashboard 1 – Brand by Body Style & EV Catalogue

Sheet 7 – Brand by Body Style

Packed-bubble chart mapping each brand to its body style category (SUV, Sedan, Hatchback, MPV, etc.), sized by model count.

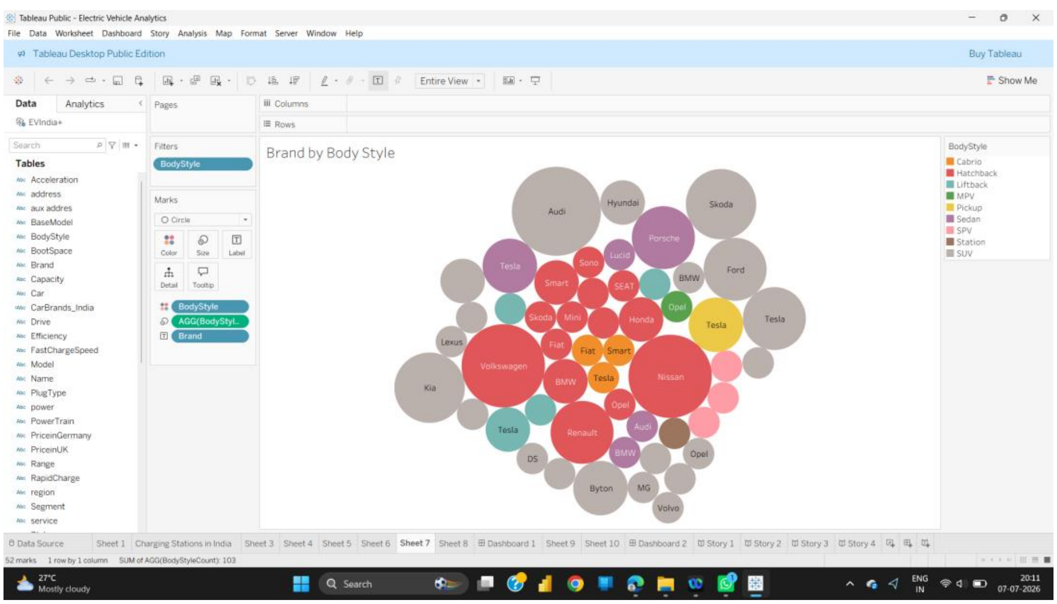


Figure: Sheet 7 – Brand by Body Style

Sheet 3 – Different EV Cars in India

Catalogue view listing all 12 Indian EV models covered in the EVIndia dataset, colour-coded by model.

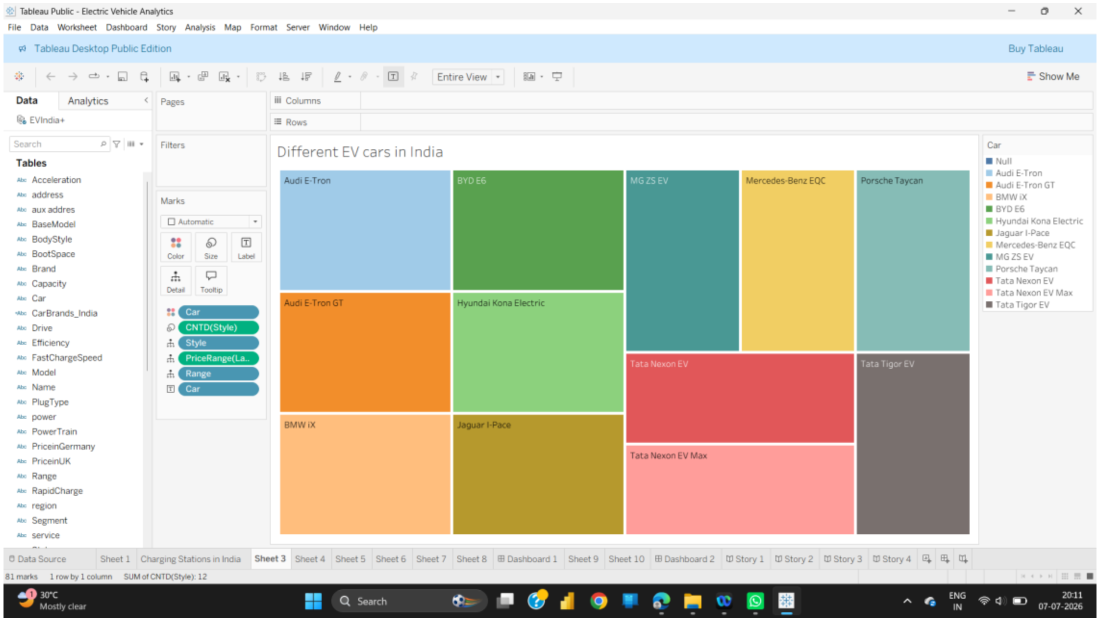


Figure: Sheet 3 – Different EV Cars in India

Dashboard 1 – Distribution of Brands & Efficiency / Top Speed

Treemap of brand distribution alongside the Top-10 Most Efficient Brands and Top Speed comparison charts.

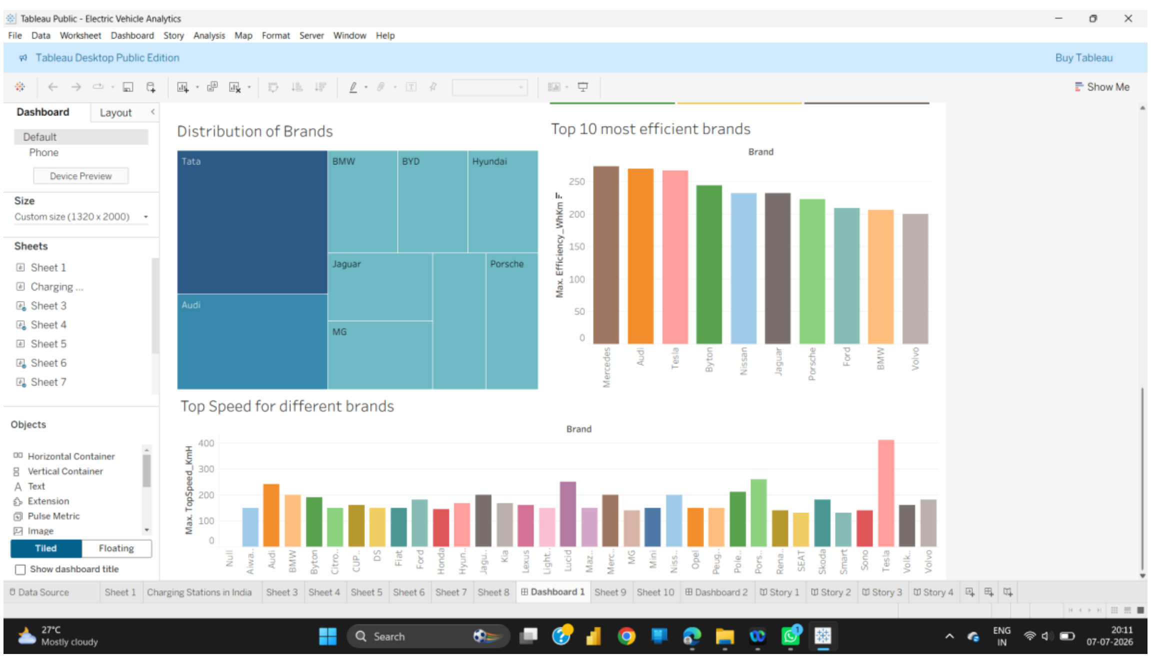


Figure: Dashboard 1 – Distribution of Brands & Efficiency / Top Speed

Sheet 8 – Distribution of Brands

Treemap showing the number of models offered by each global EV brand (Tata, Audi, BMW, BYD, Hyundai, Jaguar, Mercedes-Benz, Porsche, MG).

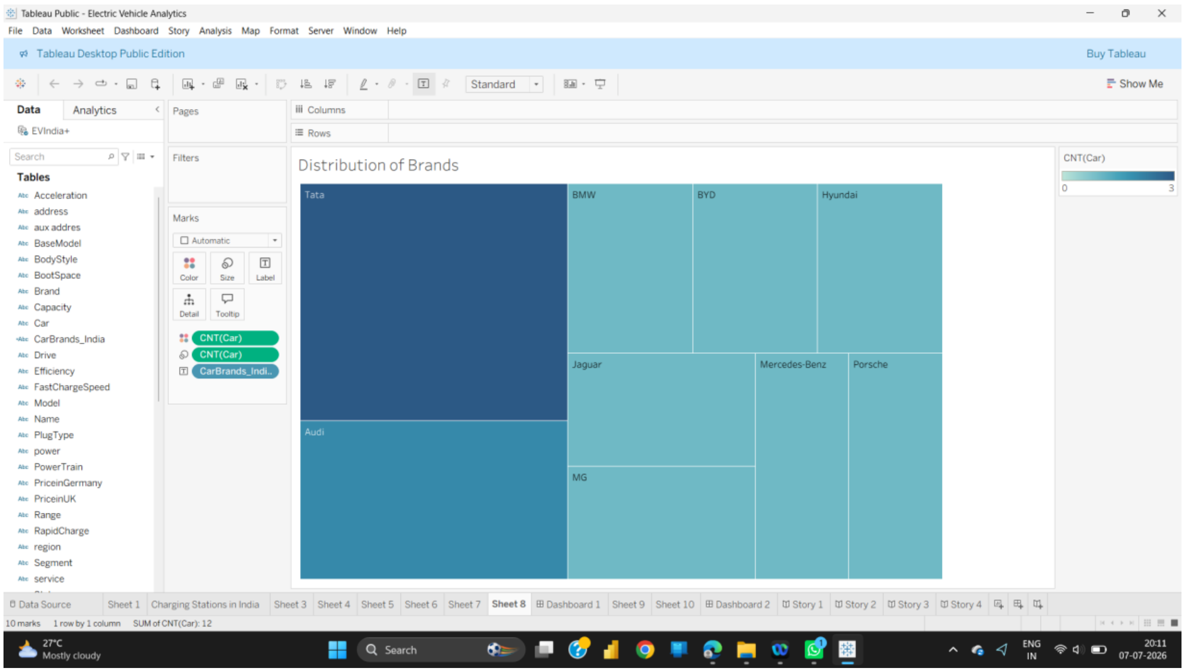


Figure: Sheet 8 – Distribution of Brands

Sheet 6 – Top 10 Most Efficient Brands

Bar chart ranking brands by maximum efficiency (Wh/km), led by Mercedes, Audi and Tesla.

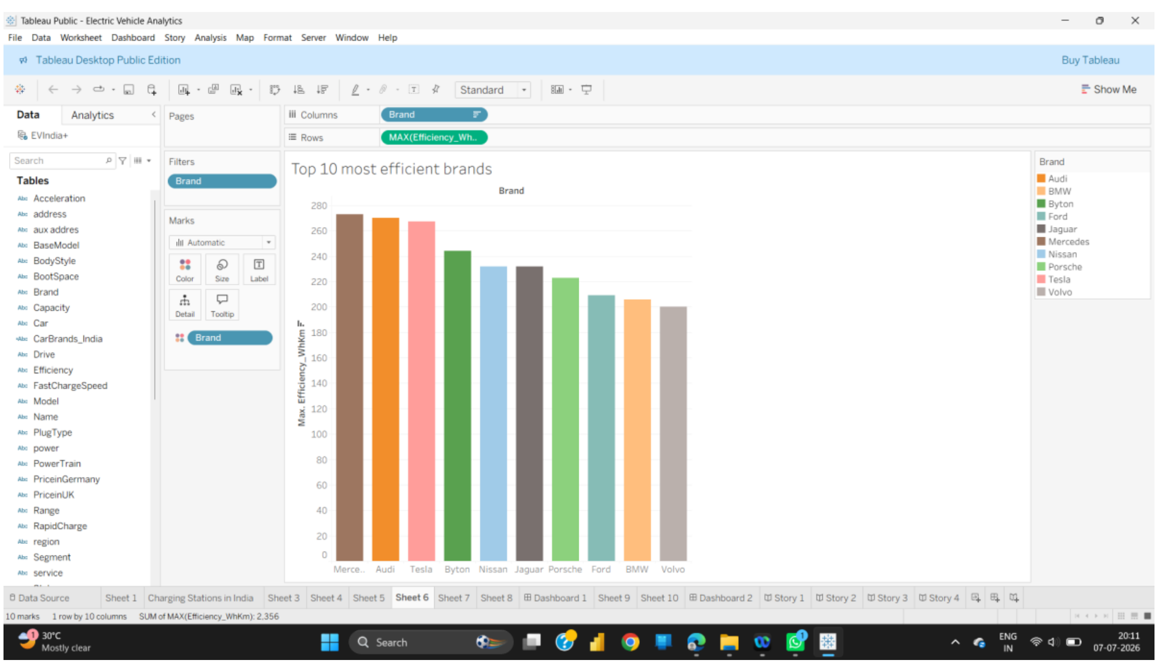


Figure: Sheet 6 – Top 10 Most Efficient Brands

Sheet 4 – Top Speed for Different Brands

Bar chart comparing the maximum top speed (km/h) across all brands in the dataset; Tesla leads at over 400 km/h.

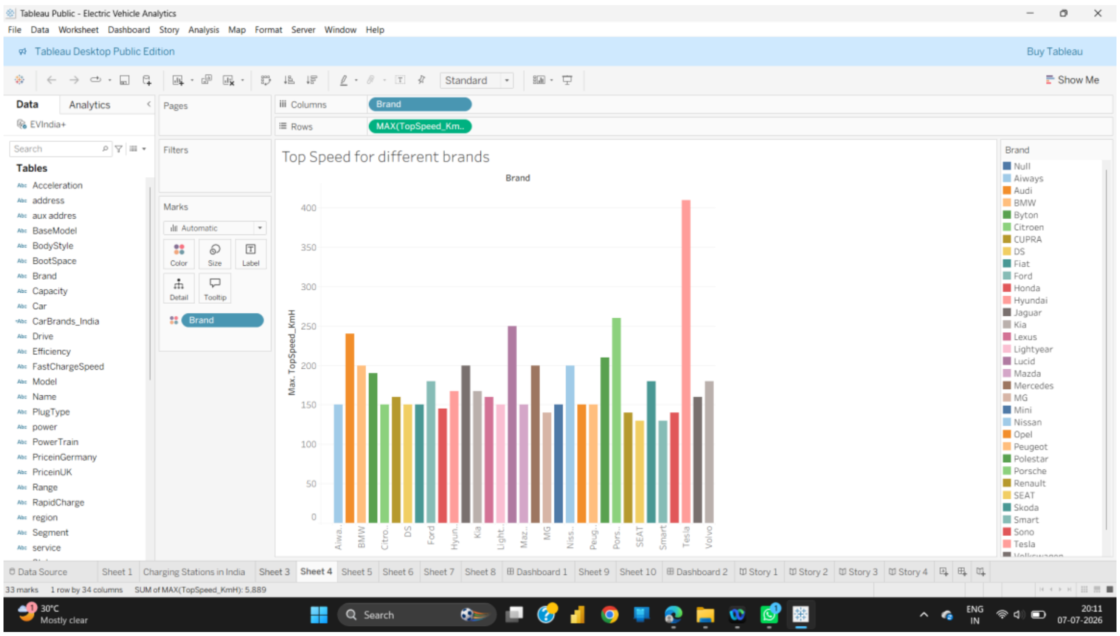


Figure: Sheet 4 – Top Speed for Different Brands

Sheet 5 – Price of Electric Cars

Horizontal bar chart of Indian EV price ranges (in Lakhs), from the Tata Tigor EV to the Audi E-Tron GT.

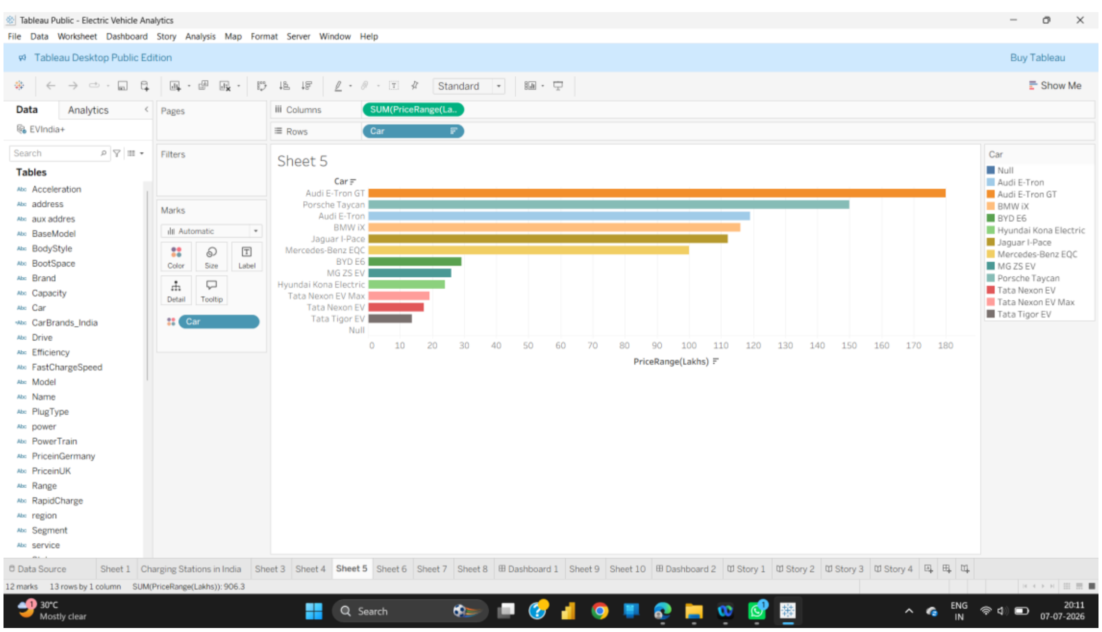


Figure: Sheet 5 – Price of Electric Cars

Charging Stations in India – Map View

Map plotting the geographic location of 202 EV charging stations across India, coloured by the managing region/authority.

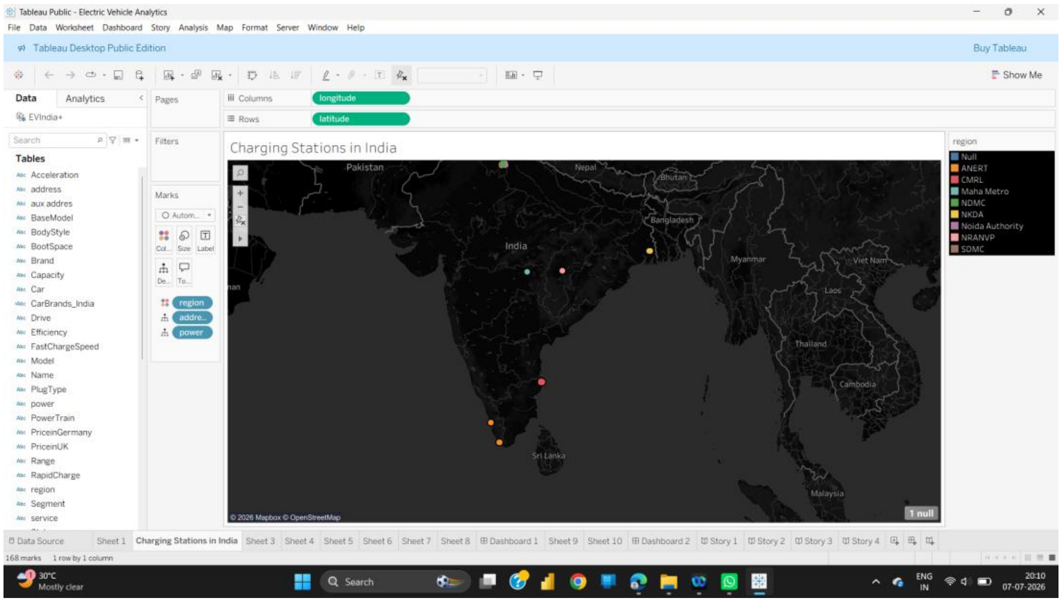


Figure: Charging Stations in India – Map View

Sheet 1 – Charging Stations by Region and Type

Stacked bar chart of charger types (AC-001, DC-001, CCS/CHAdeMO) distributed across the 8 operating regions.

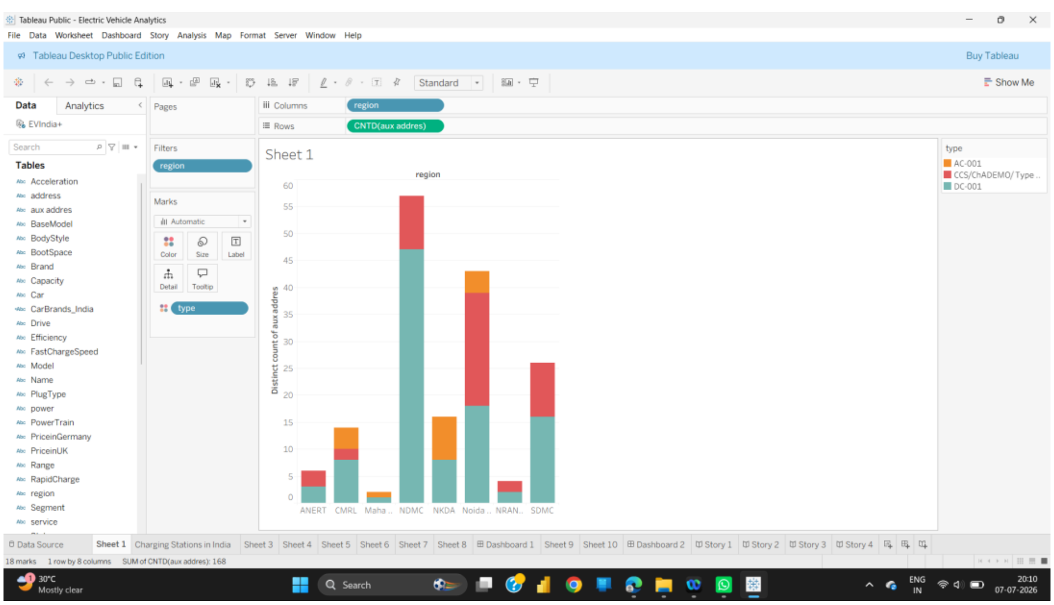


Figure: Sheet 1 – Charging Stations by Region and Type

Story 1 – Charging Stations in India

First point of the guided story introducing the charging station map to the viewer.

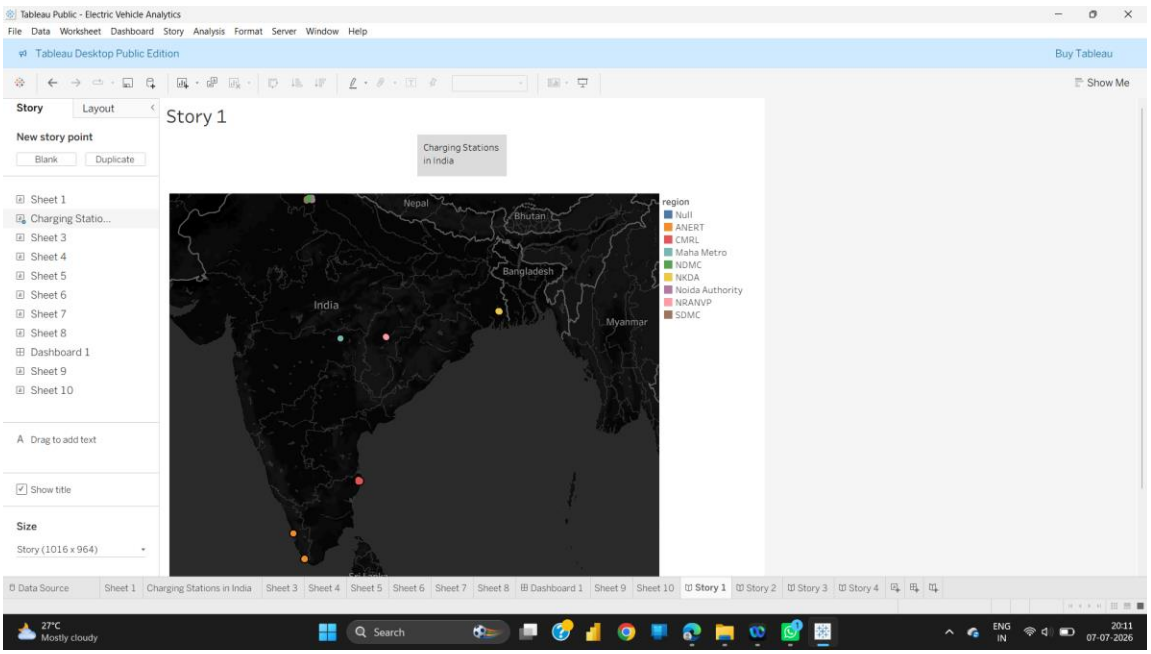


Figure: Story 1 – Charging Stations in India

Story 2 – Charging Station by Region and Type

Second story point highlighting charger-type distribution across regions.

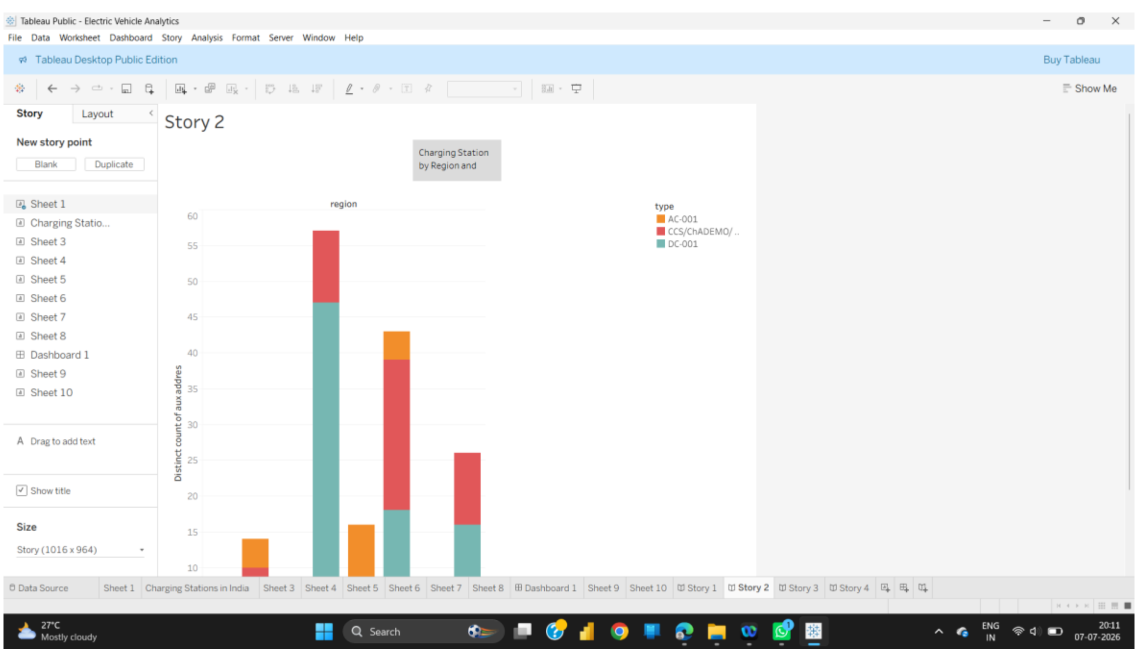


Figure: Story 2 – Charging Station by Region and Type

Story 3 – Price of Electric Cars by Different Models

Third story point walking through EV pricing across Indian models.

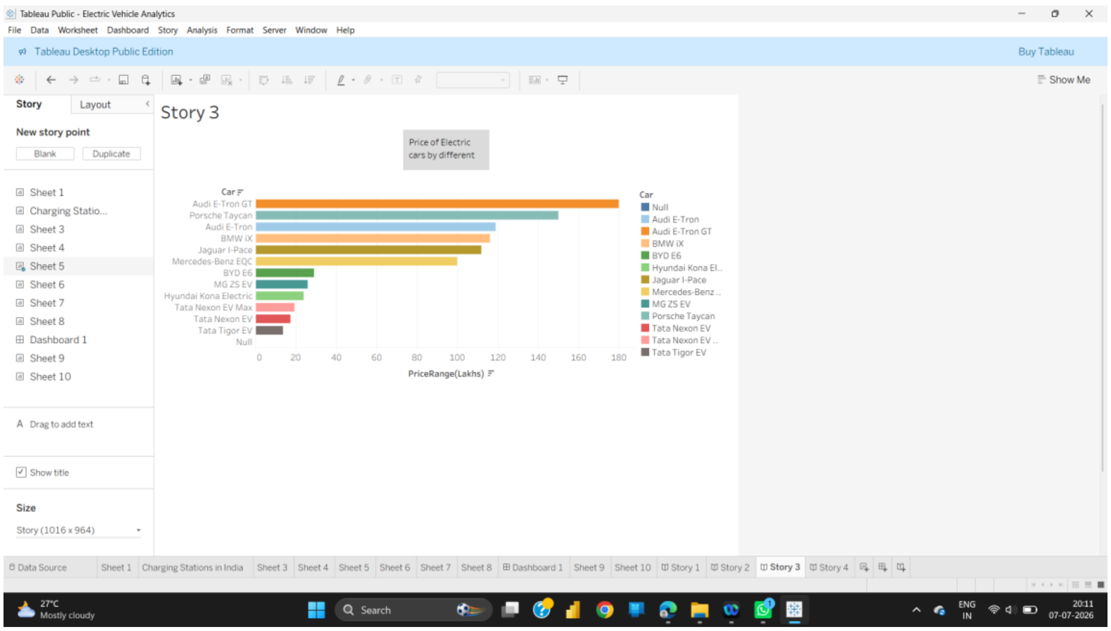


Figure: Story 3 – Price of Electric Cars by Different Models

Story 4 – Different Brands and Number of Models

Final story point summarizing brand variety and the number of models per brand.

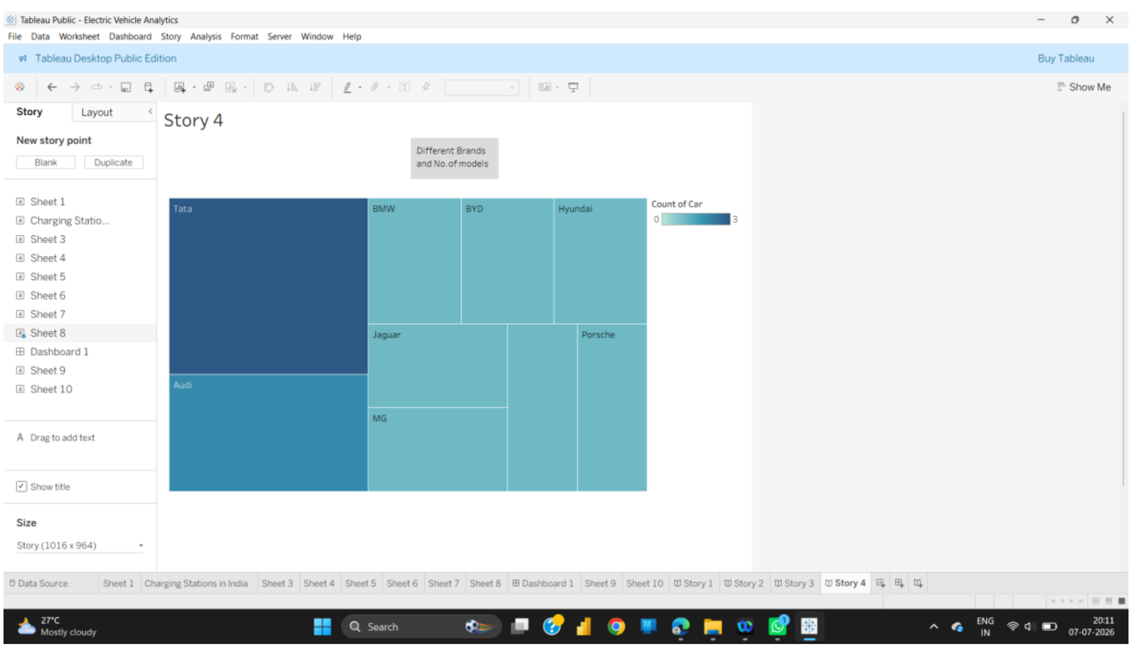


Figure: Story 4 – Different Brands and Number of Models

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides a single, consolidated view of EV specifications and charging infrastructure, eliminating the need to browse multiple sources.
- Interactive filters and story points make it easy for non-technical users to explore and compare EVs.
- Combines global (103 models) and Indian-specific (12 models) EV data, giving both a broad market view and local relevance.
- Visualizes charging station coverage across 8 regions in India, helping identify areas that are well served and areas that may need more infrastructure.
- Published on Tableau Public, making the dashboard freely and instantly accessible via a web browser without installation.
- Highly visual format (treemaps, bubble charts, maps) makes complex comparisons intuitive and quick to grasp.

Disadvantages

- The datasets are static CSV snapshots and do not update automatically as new EV models or charging stations are launched.
- Charging station data currently covers only 8 regions/operators in India, which is not an exhaustive representation of the entire country.
- Price data spans different currencies and markets (Euro, GBP, Indian Lakhs) for different datasets, requiring careful interpretation when comparing across sources.
- Tableau Public dashboards are viewable by anyone, so the underlying data is not private or access-restricted.
- Does not include real-time factors such as live charger availability, queue times, or dynamic electricity tariffs.

9. CONCLUSION

The "Electric Vehicle Charge and Range Analysis" project successfully demonstrates how data analytics and visualization can simplify decision-making in the growing electric vehicle market. By consolidating specification data for over 100 global EV models and 12 Indian EV models with charging-infrastructure data covering 202 charging stations across 8 regions in India, the project delivers actionable insights through an intuitive set of Tableau dashboards and a guided story.

The analysis reveals clear leaders in efficiency (Mercedes, Audi, Tesla) and top speed (Tesla), a wide price range in the Indian EV market (from around ₹13.6 Lakhs to ₹180 Lakhs), and an uneven but growing distribution of charging stations, concentrated around regions such as NDMC (Delhi) and SDMC (South Delhi). Overall, the project fulfils its objective of bridging the gap between vehicle-selection research and charging-infrastructure awareness, providing a valuable reference tool for prospective and current EV owners.

10. FUTURE SCOPE

- Integrate live/real-time data feeds (via API) for charging station availability and EV pricing instead of static CSV snapshots.
- Expand the charging station dataset to cover all states and Union Territories of India for a more complete infrastructure picture.
- Add predictive analytics (e.g., forecasting future charging demand or EV adoption trends) using historical data.
- Incorporate route-planning features that combine vehicle range with nearby charging station availability for trip planning.
- Include total-cost-of-ownership calculations (purchase price + running cost + charging cost) for more holistic comparisons.
- Extend the dashboard to mobile apps for on-the-go access by EV owners.

11. APPENDIX

Source Code

This project was developed using Tableau Desktop (Public Edition), a no-code/low-code visual analytics tool. There is no traditional programming source code; the 'logic' of the project consists of calculated fields, filters, and worksheet/dashboard configurations built within the Tableau workbook (Electric_Vehicle_Analytics.twbx).

Datasets:

The following datasets were used in this project (attach/host links as applicable):

- ElectricCarData_Clean.csv – Global EV specifications (103 records, 33 brands)
- EVIndia.csv – Indian EV market data (12 records)
- Cheapestelectriccars-EVDatabase.csv – Cheapest EV database (180 records)
- electric_vehicle_charging_station_list.csv – EV charging station list for India (202 records, 8 regions)

GitHub & Project Demo Link

- GitHub Repository: Poojitha-0508/Electric-vehicle-charge-and-range-analysis
- Tableau Public / Project Demo Link: Electric Vehicle Analytics | Tableau Public